Introduction to Statistics
Problem set 1

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1. In research, there was interest to evaluate the distribution of the body mass index. Hence six men and women were selected to the sample and the following data were measured from them.

Men:

height	175	180	165	190	174	191
weight	80	84	74	95	78	88

Women:

height	170	162	165	160	158	174
weight	70	62	59	55	52	66

(a) Create **SPSS** dataset from the above information. The dataset should contain variables named as height,weight, and gender. Let us denote the variable height as $Y = \text{height}$. Calculate what is the sum of the values of the variable $Y = \text{height}$ in the dataset, i.e., calculate what is the sum

$$\sum_{i=1}^{n} y_i = y_1 + y_2 + \cdots + y_n.$$

In your answer, return only the value of the sum.

(1 point)

(b) What is the sample size n in this study, i.e., how many sampling units i there are in the dataset?

(1 point)

(c) Calculate the sample median of the variable $Y = \text{height}$.

(1 piste)

(d) Create a new body mass index variable

$$\text{bmi} = \frac{\text{weight}}{(\text{height}/100) * (\text{height}/100)}.$$

Let us denote from now on the variable bmi as $Y = \text{bmi}$. Calculate the sample mean of the variable $Y = \text{bmi}$:

$$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{y_1 + y_2 + \cdots + y_n}{n}.$$

(2 points)

(e) Create a new variable $W = \text{difference}$, where the observed value w_i of the variable W is calculated as

$$w_i = y_i - \bar{y},$$

when y_i is the observed value of the variable $Y = \text{bmi}$ for the sampling unit i, and $\bar{y}$ is the sample mean of the variable $Y = \text{bmi}$. After creating the new variable, calculate the sample standard deviation of the variable $W = \text{difference}$ denoted as s_w. In your answer, return only the value of the sample standard deviation s_w.

(1 point)

2. A research on sleep quality aimed to find out how a new medication affects the sleep quality of people with sleep disorders. Persons who took the part of research were asked their own opinion on their sleep quality both before the start of the medication, the variable $Y_1 = \mathsf{start}$, and two months after the start of the medication, the variable $Y_2 = \mathsf{end}$. The self-assessment was measured by the following sleep quality score:

Very poor										Very good
0	1	2	3	4	5	6	7	8	9	10

The following data was obtained from the subjects regarding the questions:

start - Y_1	1	4	3	2	3	2	5	2	5	3	5	4
end - Y_2	4	5	7	5	4	2	4	7	6	4	8	3

Data gathered during the research is available in a file **sleepquality.sav**.

(a) Create a new variable

$$\mathsf{difference} = \mathsf{end} - \mathsf{start} = Y_2 - Y_1.$$

Then calculate the sample mean

$$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{y_1 + y_2 + \cdots + y_n}{n}$$

of the variable $Y = \mathsf{difference}$.

(2 points)

(b) Calculate the sample variance s_y^2 of the variable $Y = \mathsf{difference}$.

(1 point)

(c) Calculate the sample median of the variable $Y = \mathsf{difference}$. (1 point)

(d) Classify the values of the $Y = \mathsf{difference}$ variable into class intervals

$$(-\infty, 0], (0, \infty),$$

where $-\infty$ means minus infinite and ∞ means infinite. What is the relative frequency (percentage) of the class interval $(0, \infty)$, i.e., what percentage of the observed values of the variable $Y = \mathsf{difference}$ is between $(0, \infty)$?

(1 point)

(e) Based on the measures you have calculated above, give your preliminary conclusions of which of the following statements is most plausible.

i. The new medication has no significant (on average) effect on the personal perception of sleep quality among people with sleep disorders.

ii. The new medication significantly (on average) improves the personal perception of sleep quality among people with sleep disorders.

iii. The new medication significantly (on average) impairs the personal perception of sleep quality among people with sleep disorders.

(1 piste)

3. The company has large stores in Helsinki and Tampere. The company wanted to find the customer satisfaction of the stores. In the outlets of the stores, customers were able to express their satisfaction by the following type of survey (How satisfied were you with our customer service?):

Below is given the subset of the data. Note that the coding of the variable satisfaction is in, some sense, descending order. The file **customersurvey.sav** contains the whole data of the survey.

	location	satisfaction
1	1	2
2	1	1
3	1	4
4	1	4
5	1	1
6	1	4
.		
.		
7549	2	4
7550	2	3

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location:        satisfaction:
Helsinki = 1     Very satisfied = 1
Tampere = 2      Satisfied = 2
                 Neither satisfied nor dissatisfied = 3
                 Dissatisfied = 4
                 Very dissatisfied = 5
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(a) What kind of variable Y=satisfaction is in this study?

i. Explanatory variable defined on ordinal scale.
ii. Response variable defined on ordinal scale.
iii. Numerical explanatory variable.
iv. Numerical response variable.

(1 point)

(b) Consider the frequency distribution of the variable Y = satisfaction without considering the effect of the variable X = location on the analysis. Denote the frequency of the value $y = 1$ = "Very satisfied" by the symbol f_1. What is the value of f_1?

(2 points)

(c) Consider the frequency distribution of the variable Y = satisfaction without considering the effect of the variable X = location on the analysis. Denote the relative frequency of the value $y = 5$ = "Very dissatisfied" by the symbol p_5. What is the value of p_5?

(2 points)

(d) Consider the frequency distribution of the variable Y = satisfaction without considering the effect of the variable X = location on the analysis. Denote the cumulative relative frequency of the value $y = 4$ = "Dissatisfied" by the symbol P_4. What is the value of P_4?

(1 point)